

Internship Project • Summer 2025

SpeakShift

Zoom-integrated application for eliminating language barriers

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Agenda

- Background
- Achievements
- Demo/Functionalities
- Architecture
- Latency Optimization
- Application Evaluation

Background



Work Efficiency &
Inclusiveness



Across Entire
Organization



Foundation for Further
Development

Achievements



Full-Stack Development

Audio Processing

UI • WebSocket API



Zoom Integration

Zoom Authentication

Zoom Meetings API



Latency Reduction

Model Optimization

Service Decoupling



DevOps

CI/CD

IaC



Load Testing

Production-Like

Data Analysis

Week 1 - 2

Initial Setup
Background Research

Week 3 - 4

Architecture Finalization
Full-Stack Web MVP

Week 5 - 6

Zoom App Prototype
Multi Tenancy

Week 7 - 8

Personalized Features
Latency Reduction

Week 9 - 10

DevOps
Load Testing

Demo & Functionalities

Core Functionality

Real-Time
Transcription,
Translation,
and Voice-Over

Multi Tenancy

Personalization

Simple,
Intuitive,
User-Friendly

Multi Tenancy

Personalization

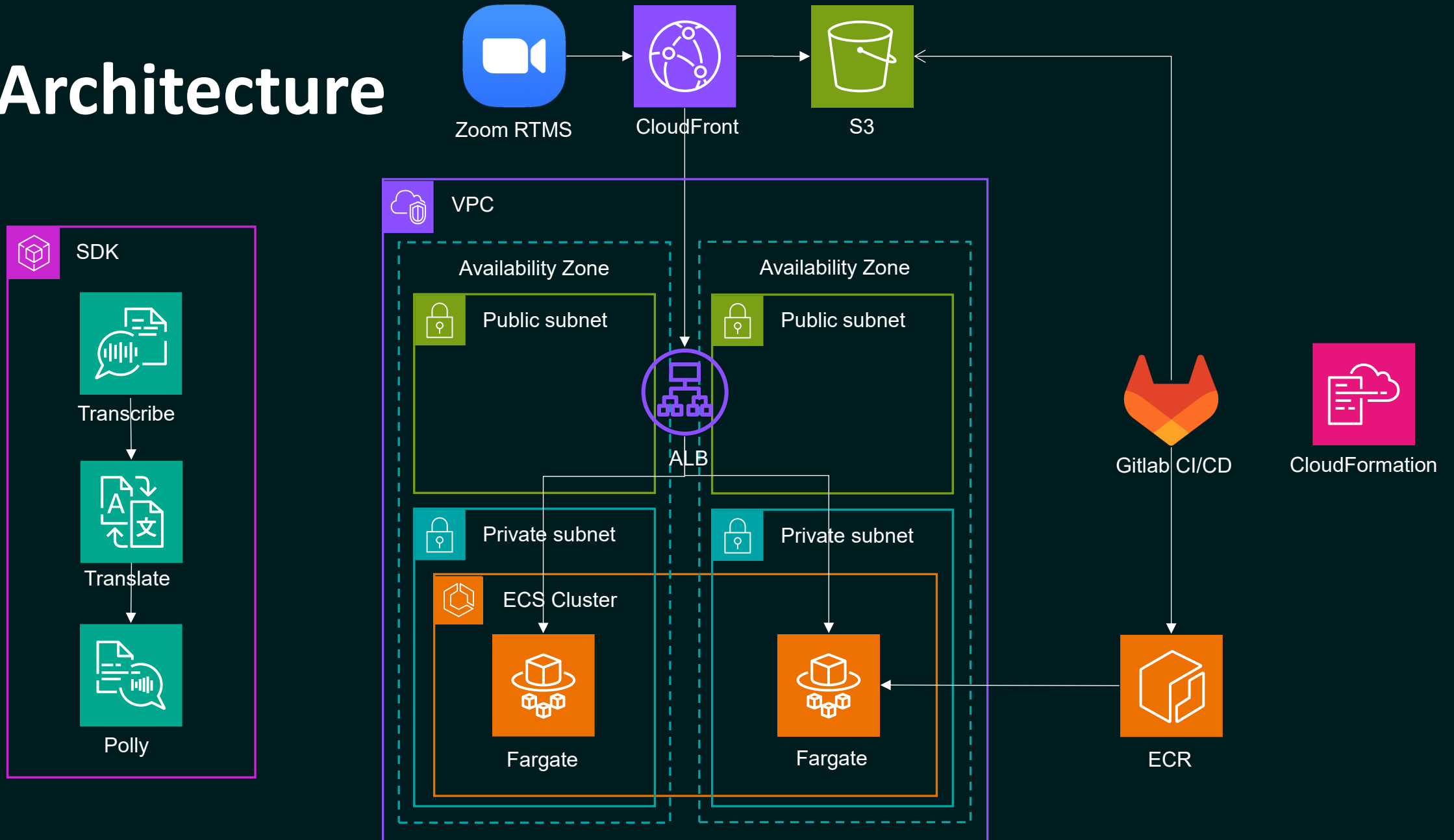
Individual
Language
Configuration

Multi Tenancy

Personalization

Broadcast
Mode for
Everyone

Architecture



Latency & Bottleneck



Architecture



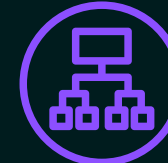
Zoom RTMS



WebSocket



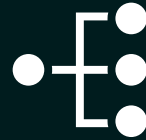
ECS Fargate



ALB



Backend



Asynchronous
Workers



Decoupling
Services



Smaller, Faster
Audio Chunks



Voiceover
Streaming



Model



Early
Finalization



Partial Data
to Frontend

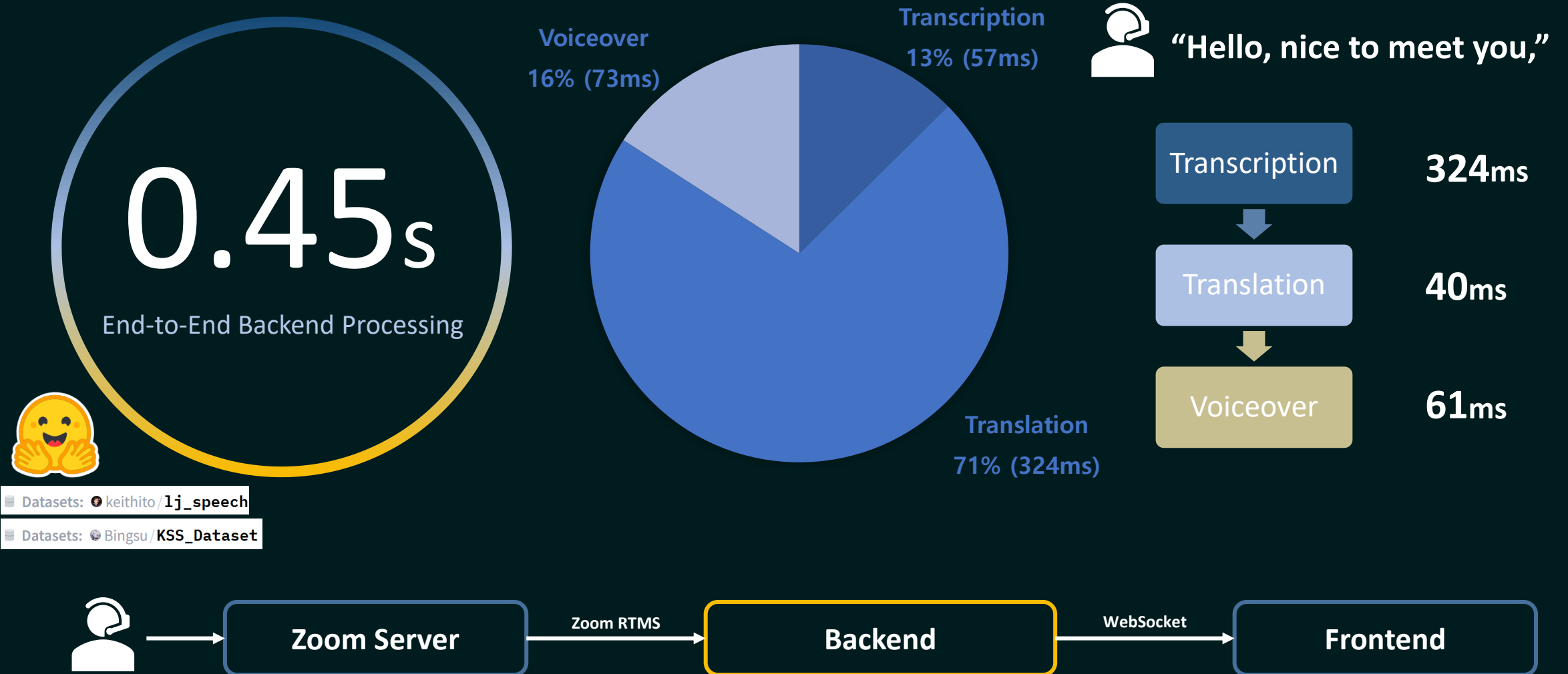


Ensure
High Stability



Model
Comparison

Application Evaluation



Load Testing



Thank you!

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